Yr 10 Yearly Marking Criteria

Section A

1.	D	A O	B O	C O	D X
2.	A	A X	B O	C O	D O
3.	A	A X	B O	C O	D O
4.	C	A O	B O	C X	D O
5.	B	A O	B X	C O	D O
6.	B	A O	B X	C O	D O
7.	C	A O	B O	C X	D O
8.	C	A O	B O	C X	D O
9.	B	A O	B X	C O	D O
10	A	A X	B O	C O	D O
11.	B	A O	B X	C O	D O
12.	D	A O	B O	C O	D X
13.	B	A O	B X	C O	D O
14.	C	A O	B O	C X	D O
15.	D	A O	B O	C O	D X
16.	C	A O	B O	C X	D O
17.	C	A O	B O	C X	D O
18.	A	A X	B O	C O	D O
19.	D	A O	B O	C O	D X
20.	A	A X	B O	C O	D O

SECTION B

Question 1

Darwin/Wallace contended that evolution is due to the natural variation that occurs in a population being acted upon by the environment. Those finches that had favorable characteristics for survival in a particular environment on an island will survive and reproduce offspring that will inherit the favorable traits for survival and so on. As the environment changed so did the types of organisms that existed.

	Marking Criteria	Marks
	1. variety existed in the finch population 2. finches that had suitable adaptations for a particular environment survived 3. reproduction of adapted finches	3
	Any two of the above	2
	Any ONE of the above	1

Question 2

The pentadactyl limbs of the mammals have the same basic structure. These similarities indicate that these organisms had common ancestry. The common ancestor had a basic limb structure that gradually changed over time due to selection pressures encountered by the different ancestral mammals.

	Marking Criteria	Marks
	1. demonstrated understanding of evolution (gradual change over time) 2. common ancestor 3. pentadactyl limbs have the same basic structure.	3
	Any two of the above	2
	Any ONE of the above	1

Question 3:

- ***Genetic variation*** *refers to differences in the genetic makeup of individuals in a population.*
- *Genetic variation is necessary in* ***natural selection****. In natural selection, organisms with environmentally selected traits are better able to adapt to the environment and pass on their genes.*
- *Major causes of variation include mutations, gene flow, and sexual reproduction.*
- ***DNA mutation*** *causes genetic variation by altering the genes of individuals in a population.*

Genetic variation is important to the processes of ***natural selection*** *and biological evolution. The genetic variations that arise in a population happen by chance, but the process of natural selection does not. Natural selection is the result of the interactions between genetic variations in a population and the environment. The environment determines which genetic variations are more favorable or better suited*

for survival. As organisms with these environmentally selected genes survive and reproduce, more favorable traits are passed on to the population as a whole.

Genetic variation occurs mainly through DNA mutation, gene flow (movement of genes from one population to another), and sexual reproduction. Due to the fact that environments are unstable, populations that are genetically variable will be able to adapt to changing situations better than those that do not contain genetic variation

A mutation is a change in the DNA sequence. These variations in gene sequences can sometimes be advantageous to an organism. Most mutations that result in genetic variation produce traits that confer neither an advantage or disadvantage. Mutations lead to genetic variation by altering genes and alleles in a population. They may impact an individual gene or an entire chromosome. Although mutations change an organism's genotype (genetic makeup), they may not necessarily change an organism's phenotype.

1. *Mutations in many bacteria that allow them to survive in the presence of antibiotic drugs. The mutations lead to* ***antibiotic-resistant*** *strains of bacteria.*
2. *Sick cell anaemia develops resistance to Malaria*
3. *Peppered moth -Variation*
4. *Finches- for Variation*
5. *Rock pocket mouse- Mutation.*

Marking Criteria	Marks
- Clear differentiation between Mutation and Variation. - Linking Variation with Natural Selection - Statement on how mutation and variation play role in species success - Example for either mutation or variation.	4
Any 3 of the above	3
Any TWO of the above	2
Any One of the above	1

Teacher comments:

Most of the students were Confused between mutation and variation. Some failed to identify the link between variation and selection.

Clear examples not provided (most of them were vague and expanding on the examples is not done clearly)

Random general examples did not receive marks e.g.: if suddenly some catastrophe occurs/temperature increases etc. it helps species to survive. OR some disease spreads it prevents the species from dying.

Question 4

Marking Criteria	Marks
Name ONE identified biotechnology Outline the biotechnology. 2 valid points in support 2 valid points in opposing	4
Name ONE identified biotechnology Outline the biotechnology. 1 valid points in support 1 valid points in opposing **(OR)** Outline the biotechnology. 1 valid points in support 1 valid points in opposing	3
Name ONE identified biotechnology 1 valid points in support 1 valid points in opposing **(OR)** Name ONE identified biotechnology 1 valid points in support with general info on points for opposition.	2
Relevant information	1

Teachers Comment:

Common misunderstanding

GMO: is a product or more commonly used term for genetically modified (GM), as it is commonly used, refers to the transfer of genes between organisms using a series of laboratory techniques for cloning genes, splicing DNA segments together, and inserting genes into cells. Collectively, these techniques are known as ***recombinant DNA technology****.*

GMO ITSELF IS NOT A TECHNOLOGY

General information points like: it increases quality of crops is not considered as a support point. What quality is being referred to? (Nutritional, meat content, shell life etc.)

Be careful when talking about cloning. Cloning at this stage is not successful to create an entire organism. Cloning of some specific traits (somatic transfer) can be introduced into ovum to which then has to be implanted into the womb and then will undergo normal developmental process. This process has seen some success in Animal husbandry/selective breeding. Any cloning related to human beings is not allowed.

Therapeutic cloning is done for medical purposes in labs. No ethical issues are involved in it.

STEM CELL Therapy is the name of the technology. Stem cells cannot be created without prior seed / stem cell potent material. It is incorrect if you write stem cells with desired characters are made (stem cells from the target or desired organism are Cultured under ideal conditions would be more appropriate) and inserted to human being to get gain immunity/ get rid of diseases.

stem cell transplants, stem cells replace cells damaged by chemotherapy or disease or serve as a way for the donor's immune system to fight some types of cancer and blood-related diseases, such as leukemia, lymphoma, neuroblastoma, and multiple myeloma.

Question 5

(a) Beth is heterozygous as one of his children do not exhibit the trait, but the other two children do. This proves that Beth carries a dominant allele and is heterozygous for the trait in contrast to his wife who is trait free homozygous recessive.

Marking Criteria	Marks
Identification of heterozygous condition Justification provided	2
Identification of heterozygous condition	1

(b) Pp,Pp,pp,pp

Marking Criteria	Marks
Correctly identifies 4 genotypes or writes Pp and pp	2
Correctly write one genotype	1

(c)

Marking Criteria	Marks
Correctly draws a punnet square using heterozygous and homozygous genotypes States that 50 % of the offspring has the condition	2
Correct Punnet square OR Correct percentage	1

SECTION C

Question 1 (3 marks)

Hydrochloric acid was added to identical masses of sodium carbonate crystals in 4 separate test tubes. The test tubes had equipment attached to measure the volume of gas produced.

The results for each test tube varied, producing 38, 35, 25, and 8mL of gas respectively after 3 minutes.

(a) Suggest one possible variable that may have caused the acid to react at different rates with the four samples of sodium carbonate. **1**

- Concentration of HCl
- Surface area of sodium carbonate crystals (agitating the test tube would result in more surface area becoming available faster)
- Different temperatures of the test tubes / HCl

Marking criteria	**Marks**
• Correctly identifies one of the above variables	**1**

The amount of HCl would not change the rate of reaction (as long as there is enough HCl to cover the crystals).

(b) Using your response to (a), suggest TWO ways that variables should have been controlled in this experiment to make it valid. **2**

- Keep concentration constant by using HCl from the same bottle/mixture, check HCl beforehand by measuring the pH
- Keep temperature constant by performing in the same room/same time, avoid sources of heat, preheat acid to a known temperature before beginning experiment.
- Keep surface area constant by crushing the crystals or by sieving the crystals, use the same source/supplier/style of crystals, use solid blocks of sodium carbonate crystals, either mix all test tubes or do not mix any of them.

Marking criteria	**Marks**
• Correctly identifies 2 ways that the variable chosen in (a) could be kept constant • or • Correctly identifies 2 of the above variables and reasonable ways they could be controlled	**2**
• Correctly identifies one of the above variables and a reasonable way to control it or • Correctly identifies 2 of the above variables	**1**

Some students mentioned variables that needed to be kept constant but did not suggest ways this could be achieved

Question 2 (9 marks)

A student was asked to investigate the effect of temperature on the time taken for a piece of magnesium to dissolve in hydrochloric acid.

(a) She used a test tube, strips of magnesium, and 1M hydrochloric acid. List THREE other pieces of equipment she would need to use. **1**

Most important pieces of missing equipment:

- Bunsen burner / hotplate
- Thermometer
- Stopwatch / phone - to time the reaction
- Refrigerator / Ice bath
- Measuring cylinder

Other accepted pieces of equipment:

- Safety glasses
- Retort stand + clamp to hold test tube and or thermometer / tongs
- Beaker (to fill with water to heat more gently or to fill with ice water to cool the test tube)
- Test tube rack
- Pipette / dropper
- Ruler to measure magnesium strips

Marking criteria	**Marks**
• Correctly identifies 3 of the above pieces of equipment	**1**

Students needed to list specific pieces of equipment, heating apparatus etc was not accepted.
Generic equipment such as personal PPE & safety glasses etc were not accepted.
NOTE – Safety glasses was initially not accepted but this was changed, please let your teacher know if you did not get the mark for it.

(b) She used 5 different temperatures and timed 3 trials for each temperature. An extract from her notebook is shown below.

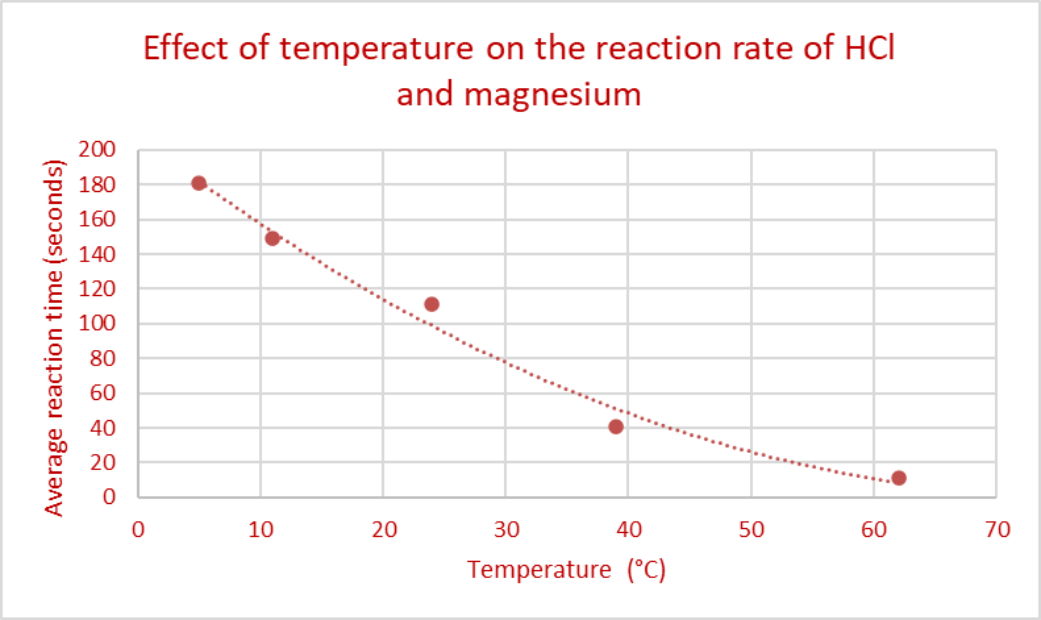
5°C 180 secs 171 secs 199 secs	11°C 158 secs 140 secs 149 secs	24°C 98 secs 102 secs 133 secs	39°C 50 secs 38 secs 35 secs	62°C 11 secs 9 secs 13 secs

Present these results in a table and calculate the average reaction time for each temperature. **3**

Temperature (°C)	**Reaction time (seconds / s)**			
	Trial 1	**Trial 2**	**Trial 3**	**Average**
5	180	171	199	183 (183.3)
11	158	140	149	149
24	98	102	133	111
39	50	38	35	41
62	11	9	13	11

Marking criteria	Marks
• Table with data and appropriate headings (within the table) • Units in headings not with the data (seconds or s accepted) • Independent variable on the left column/top row & all headings in the top rows. • Averages correctly calculated	**3**
• Provides 2-3 of the above	**2**
• Only 1 of the above	**1**

(c) Graph the average results on the grid below **3**



Marking criteria	Marks
• Correct type of graph (scatter plot) • Correct Axes (x temp, y time) • Appropriately scaled • Correct plot (of student's own averages) • Curve of best fit • Axes labelled with units (Units only on labels)	**3**
• At least 4 of the above	**2**
• At least 2 of the above	**1**

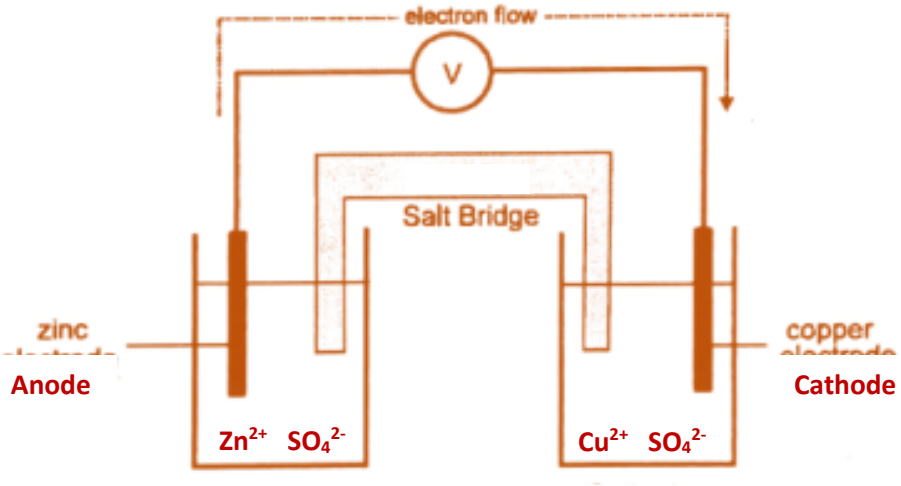
The curve of best fit does not need to be straight. The relationship is not necessarily linear, however a straight line of best fit was accepted if it intersected the data points evenly. It should not directly link between points (connect the dots).

Appropriately scaled means that the axes increase linearly and that at least half of the available space is used by the graph.

Question 3 (4 marks)

Draw a labelled diagram of a galvanic cell with a voltmeter, zinc and copper electrodes, a salt bridge, and zinc sulfate and copper sulfate as the electrolytes. Also label the anode, cathode, salt bridge, and direction of electron flow.

The half equations for the cell are: is $Zn \rightarrow Zn^{2+} + 2e^-$ and $Cu^{2+} \rightarrow 2e^- + Cu$.



Marking criteria	Marks
• Voltmeter location and symbol correct • Electrodes in correct location • Cathode and anode labelled correctly (copper cathode, zinc anode) • Salt bridge labelled and in correct location • Electrolytes labelled in correct location (Copper sulfate and $CuSO_4$ both accepted) • Correct direction of electron flow (towards copper cathode)	**4**
• 4-5 of the above	**3**
• 3 of the above	**2**
• 1-2 of the above	**1**

A rectangle with voltmeter written inside was also accepted, however incorrect circuit diagram symbols were not accepted.

I have not been pedantic about scientific diagram conventions, but for reference all diagrams should be 2D, and objects should be labelled with a straight line touching the object, not arrows.

Question 4 (2 marks)

Solutions of aluminium bromide and potassium sulfate are mixed to form potassium bromide and aluminium sulfate. No precipitate is formed.

Write a balanced chemical equation with states for the above reaction. **2**

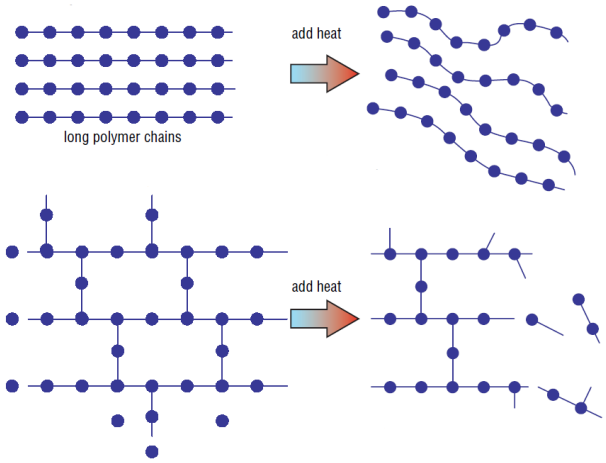
$2AlBr_{3(aq)} + 3K_2SO_{4(aq)} \rightarrow 6KBr_{(aq)} + Al_2(SO_4)_{3(aq)}$

Marking criteria	Marks
• Correct formulas for products and reactants • Balanced equation • Correct states included	**2**

• Adheres to chemical equation conventions (reactants on left, products on right, plus and arrow symbols, correct size of coefficient and subscript numbers)	
• Provides at least 2 of the above	**1**

Question 5 (4 marks)

The following diagram shows the structures of 2 types of plastic polymers and what happens when heat is applied to them.



Identify the 2 types of plastics and explain what happens to each one when burned.

The top polymer with long chains is a thermoplastic which melts when heated, as heat causes the long chains to slip over each other.

The bottom polymer with cross-linked chains is a thermosetting plastic which will char when heated/burnt but will not soften or melt.

Marking criteria	**Marks**
• Correctly identifies the 2 types of plastics and explains the effects of burning them	**4**
• Correctly identifies the 2 types of plastics and identifies the effects of burning one of them	**3**
• Correctly identifies the 2 plastics OR • Correctly identifies 1 plastic and the effect of burning it OR • Correctly explains what happens to each plastic when burned	**2**
• Correctly identifies 1 type of plastic OR • Correctly explains what happens to one type of plastic when burned.	**1**

Marks were not given to descriptions that were only focused on what happens at the atomic level as shown in the images. You had to mention that the plastic becomes malleable, not that the polymer chains become malleable.

SECTION D

Question 1 (6 marks)

(a) Explain the difference between AC and DC electricity, giving examples of where each one is used. **2**

Marking Criteria	*Mark(s)*
- Explains the difference between direction of current in AC and DC electricity - Gives an example for each	**2**
1 of the above	**1**

AC electricity uses alternating current, which means the current changes direction periodically. This is used in power lines.
DC electricity uses direct current, which means the current only flows in one direction. This is used in batteries.
Notes- need to specify changing direction. 'Alternating' or 'fluctuating' is not enough, could mean fluctuating in amount rather than direction.

(b) Electricity is supplied in Australian households at 240V. However, electric power is transported over large distances at high voltages (e.g. 66kV) rather than at 240V. Explain why. **1**

Marking Criteria	*Mark(s)*
Explains that higher voltages means less electricity is lost in transmission, saving electricity and hence money	**1**

Power lines transport electricity over long distances which means energy is lost along the way. Having high voltages means lower current which means less electricity is lost, saving money.

(c) Explain the functions of a transformer. **2**

Marking Criteria	*Mark(s)*
- Explains that transformers can be used to increase voltage - Explains that transformers can be used to decrease voltage	**2**
One of the above OR Both of the above but refers to current instead of/as well as voltage	**1**

Step-up and step-down transformers are used to increase or decrease voltages respectively.
Notes- current and voltage are different things. Saying 'transformers increase the voltage of a current' is like saying 'eating food increases the weight of your height'.

(d) Name a device used at home or in a school that uses an electromagnet. **1**

Marking Criteria	*Mark(s)*
Correctly identifies a device used in the school or home that uses an electromagnet	**1**

Speakers, headphones, school bell

Question 2 (8 marks)

(a) The Sun is a relatively small star. Sort the following stages of the Sun's life into the correct order. **2**

Stages in the life cycle of our Sun: Nebula, white dwarf, red giant and protostar.

Marking Criteria	*Mark(s)*
Correctly orders the stages in the life cycle of our Sun	**2**
One stage out of order	**1**

Nebula, protostar, red giant, white dwarf
Notes – many students confused a nebula with a planetary nebula, they're not the same

(b) The two brightest stars of the Southern Cross are Alpha Crucis and Beta Crucis. Jane told John that if they are the same brightness that means they are the same distance away from Earth.

Is Jane correct? Explain your answer. **2**

Marking Criteria	*Mark(s)*
Explains why Jane is incorrect	**2**
Identifies that Jane is incorrect	**1**

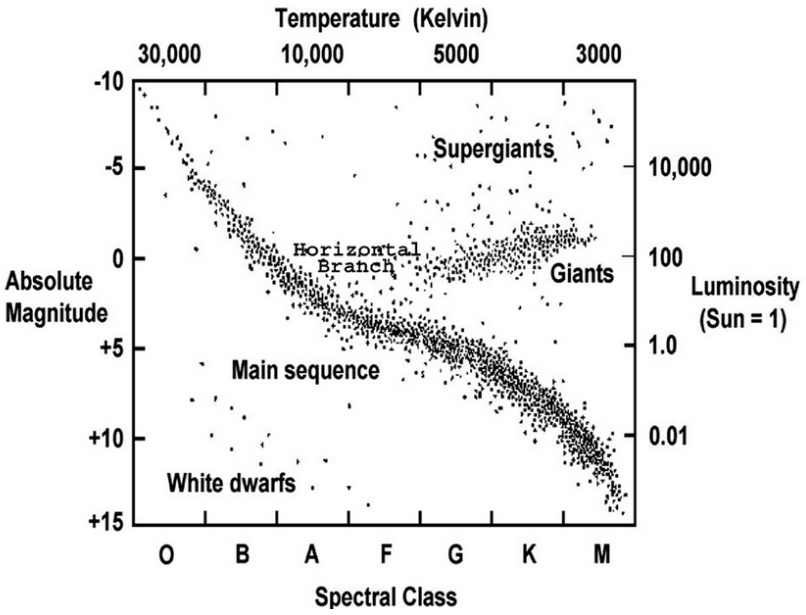
The brightness of stars when viewed on Earth depends on both how luminous the star actually is and how far away it is from us. The chances of the 2 stars being the same distance from us is extremely small, and so Jane is most likely incorrect as one star is probably brighter but further away. Notes- Only explaining what factors affect brightness does not explain why Jane is wrong. If I say standing on Earth we are equally pulled to the Sun as to Earth and you say 'gravity depends on mass and distance' that doesn't explain why I'm wrong.

(c) Identify the name of the 'explosion' that occurs when a large star dies, and the 2 possible outcomes of such an explosion. **2**

Marking Criteria	*Mark(s)*
- Correctly identifies 'supernova' - Correctly identifies 'neutron star' and 'black hole'	**2**
One of the above	**1**

A massive star dying leads to a supernova, which can result in either a neutron star or black hole depending on the star's mass

(d) The graph below plots stars according to their temperature, luminosity, magnitude and spectral class.



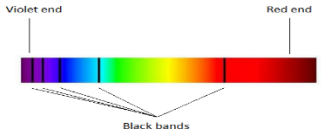
Use the graph to compare the properties of a white dwarf and a giant star. **2**

Marking Criteria	*Mark(s)*
- Identifies that white dwarves are hotter than giants - Identifies that giants are brighter than white dwarves	**2**
One of the above	**1**

White dwarves are hotter than giants on average, while giants are brighter than white dwarves on average.

Question 3 (6 marks)

(a) Identify what the following image is an example of. **1**



Marking Criteria	*Mark(s)*
Correctly identifies an absorption spectrum	**1**

Absorption spectrum

(b) Name a technology, besides the optical telescope and spacecraft, that has enhanced our understanding of the universe. **1**

Marking Criteria	*Mark(s)*
Names a technology used in space research	**1**

Rovers, charge coupled devices, spectrometer, radio telescopes, LHC, etc.

(b) Outline the Big Bang theory and explain the evidence in support of it. **4**

Marking Criteria	*Mark(s)*
Outlines the Big Bang theory with reference to the expansion of all matter and energy from a single point. Provides a detailed explanation of the Doppler effect, red shift of absorption spectra, and CMBR as evidence for the Big Bang theory.	**4**
Outlines the Big Bang theory with a basic outline of the evidence in support of it.	**3**
Outlines the Big Bang theory, describing one piece of evidence in support of it. OR Describes two pieces of evidence that support the Big Bang theory	**2**
Outlines the Big Bang theory OR Identifies a piece of evidence that supports the Big Bang theory	**1**

The 'Big Bang' theory (BBT) is the idea that the universe started from a single densely packed point that expanded into all of the energy, matter and antimatter that makes up the universe. The universe is still expanding from this initial explosion today. Evidence to support the BBT includes redshift and the doppler effect and the cosmic microwave background (CMB or CMBR).

The Doppler effect is the change in the frequency of a wave (in astronomy it is electromagnetic radiation in the form of light) as it moves towards or away from the observer (us on earth). With light this means the colour is shifted either towards the blue or red end of the spectrum – red if it is moving away from us, blue if it is moving towards us. As galaxies and other celestial objects continue to move in the expanding universe, the spectra we observe from them is red-shifted as they move away from us. This is a piece of evidence to suggest that the universe is still expanding from its original single point.

CMBR is a form of electromagnetic radiation that fills the universe and is a remnant of the early stages of the universe (which is why it is often called "relic radiation"). CMBR can be detected by special instruments in every direction of the universe and can be used to date the age of the universe and its composition.